

Reverse Outline Assignment Andrew Koren

This is a paragraph-by-paragraph reverse outline of Energy-Level Quantization in the Zero-Voltage State of a Current-Biased Josephson Junction, by John M. Marinis et al.

Original Abstract:

We report the first observation of quantized energy levels for a macroscopic variable, namely the phase difference across a current-biased Josephson junction in its zero-voltage state. The position of these energy levels is in quantitative agreement with a quantum mechanical calculation based on parameters of the junction that are measured in the classical regime.

Introduction

- We want to show quantum mechanics has an affect on large-scale systems (macroscopic variables)
- Current flow through josephson tunnel junctions depends on circuit phase difference between each side
- Past experiments have only measured tunneling rates
- We test and observe quantized energy levels by irradiating the junction with light of various frequencies
- The experiment can be represented as an RC circuit with tunnel junction
- List circuit element parameters in terms of base values $I_0, \Phi_0, \omega_{p0}, C$
- Fig 1(a) diagrams cubic potential of circuit & quantum energy levels

Device/Experiment Setup

- Tunnel junction made via photolithography
- ^4He temperature maintained on circuit
- Low-temp escape rate (tunneling rate) match theory

Note, switching rate is used to measure tunneling/escape rate

- Escape rate is measured and controlled by microwave power and bias current
- Microwave power tuned to keep escape rate with power on less than 3x escape rate with power off
- Tunneling kept low ($30\mu\text{s}$ between events) to allow cooling to equilibrium

Experimental Results

- energy level are raised/lowered above/below microwave frequency energy
- Fig 2-4: current vs switching rate with microwave power on/off
 - captions include parameters I_0, C etc.

- resonances correspond to excitation to various excited states
- match with numerical solutions to Schrödinger equation, strong agreement with results
- energy levels plotted in fig 2-4
- errors in measurement of I_0, C will not limit switching frequency measurement accuracy
- Test tunneling properties of junction with different I_0, C to compare
- Also matches predictions very well, within error bounds
- Compare RC +tunneling resonance frequencies with plasma frequency
 - does not overlap
- Quantum/Classical regimes observed via temperature variation
- Higher temperature allows for electrons with varied energies, so bands are smeared
- Not clear how damping RC is related to energy level smearing
 - appears related to $\Delta\omega/\omega = 1/Q$
 - Q measured in classical limit

Summary

- microwaves at energy levels increase tunneling
- great agreement with predictions, theory
- coherence between ground/excited state as much as 40ns

credits, citations